

LAB NOC 02

Cisco Catalyst 2960 — Port Security & MAC Address Control

Real Hardware | ES / EN | NOC Portfolio

INFORMACION DEL LABORATORIO / LAB INFORMATION

Hardware / Hardware	Cisco Catalyst 2960 (WS-C2960-24TT-L)	Cisco Catalyst 2960 (WS-C2960-24TT-L)
Categoría / Category	Seguridad de Capa 2	Layer 2 Security
Nivel / Level	CCNA — Switching / Seguridad	CCNA — Switching / Security
Duración / Duration	Aprox. 40 minutos	Approx. 40 minutes
Entorno / Environment	Hardware real (equipo usado)	Real hardware (used equipment)
Relevancia NOC	Control de acceso L2, detección de intrusión	L2 access control, intrusion detection

OBJETIVO / OBJECTIVE

OBJETIVO	OBJECTIVE
Simular un escenario real de NOC: configurar seguridad en un puerto Fast Ethernet del Catalyst 2960, detectar una posible intrusión por MAC no autorizada, y validar el comportamiento del switch (aprendizaje sticky, modo shutdown) en hardware real.	Simulate a real NOC scenario: configure security on a Fast Ethernet port of the Catalyst 2960, detect a potential intrusion via unauthorized MAC, and validate switch behavior (sticky learning, shutdown mode) on real hardware.

PROCEDIMIENTO — 3 ETAPAS / PROCEDURE — 3 STAGES

#	ESPAÑOL	ENGLISH
1	ETAPA 1 — Baseline: verificar estado inicial del switch	STAGE 1 — Baseline: verify initial switch state
2	Ejecutar: show mac address-table	Run: show mac address-table
3	Confirmar: MAC c4da.2604.8923 aprendida como DYNAMIC en VLAN 1, puerto Fa0/1	Confirm: MAC c4da.2604.8923 learned as DYNAMIC on VLAN 1, port Fa0/1
4	Documentar estado normal — clave NOC: verificar baseline antes de cualquier cambio	Document normal state — NOC key: verify baseline before any change
5	ETAPA 2 — Ingresar: interface FastEthernet 0/1	STAGE 2 — Enter: interface FastEthernet 0/1
6	Activar: switchport port-security	Enable: switchport port-security
7	Limitar a 1 MAC: switchport port-security	Limit to 1 MAC: switchport port-security

	maximum 1	maximum 1
8	Sticky: switchport port-security mac-address sticky	Sticky: switchport port-security mac-address sticky
9	Violation: switchport port-security violation shutdown	Violation: switchport port-security violation shutdown
10	Persistir: end -> write memory	Persist: end -> write memory
11	ETAPA 3 — Verificar: show port-security interface Fa0/1	STAGE 3 — Verify: show port-security interface Fa0/1
12	Confirmar todos los campos clave en la salida del comando	Confirm all key fields in command output

```

COM5 - PuTTY

--- System Configuration Dialog ---

Enable secret warning
-----
In order to access the device manager, an enable secret is required
If you enter the initial configuration dialog, you will be prompted for the enable secret
If you choose not to enter the initial configuration dialog, or if you exit setup without setting the enable secret, please set an enable secret using the following CLI in configuration mode-
enable secret 0 <cleartext password>
-----
Would you like to enter the initial configuration dialog? [yes/no]: no

```

Switch 2960 Baseline

```

COM5 - PuTTY

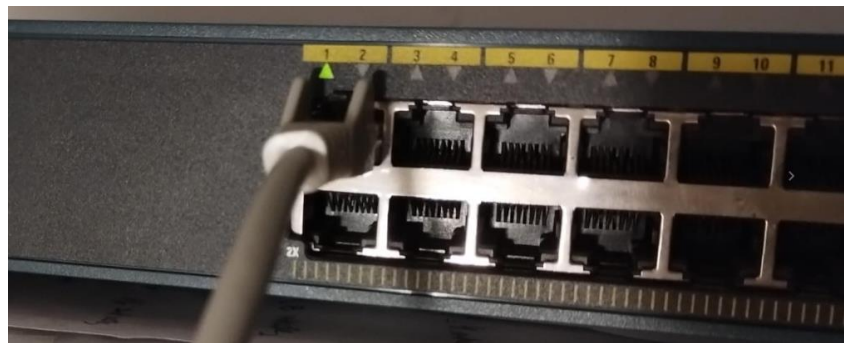
--- System Configuration Dialog ---

Enable secret warning
-----
In order to access the device manager, an enable secret is required
If you enter the initial configuration dialog, you will be prompted for the enable secret
If you choose not to enter the initial configuration dialog, or if you exit setup without setting the enable secret, please set an enable secret using the following CLI in configuration mode-
enable secret 0 <cleartext password>
-----
Would you like to enter the initial configuration dialog? [yes/no]: no
Switch>enable
Switch#

```

```
COM5 - PuTTY
-----
Would you like to enter the initial configuration dialog? [yes/no]: no
Switch>enable
Switch#show interface status

Port      Name      Status      Vlan    Duplex  Speed  Type
Fa0/1     Fa0/1     notconnect  1       auto    auto   10/100BaseTX
Fa0/2     Fa0/2     notconnect  1       auto    auto   10/100BaseTX
Fa0/3     Fa0/3     notconnect  1       auto    auto   10/100BaseTX
Fa0/4     Fa0/4     notconnect  1       auto    auto   10/100BaseTX
Fa0/5     Fa0/5     notconnect  1       auto    auto   10/100BaseTX
Fa0/6     Fa0/6     notconnect  1       auto    auto   10/100BaseTX
Fa0/7     Fa0/7     notconnect  1       auto    auto   10/100BaseTX
Fa0/8     Fa0/8     notconnect  1       auto    auto   10/100BaseTX
Fa0/9     Fa0/9     notconnect  1       auto    auto   10/100BaseTX
Fa0/10    Fa0/10    notconnect  1       auto    auto   10/100BaseTX
Fa0/11    Fa0/11    notconnect  1       auto    auto   10/100BaseTX
Fa0/12    Fa0/12    notconnect  1       auto    auto   10/100BaseTX
Fa0/13    Fa0/13    notconnect  1       auto    auto   10/100BaseTX
Fa0/14    Fa0/14    notconnect  1       auto    auto   10/100BaseTX
Fa0/15    Fa0/15    notconnect  1       auto    auto   10/100BaseTX
Fa0/16    Fa0/16    notconnect  1       auto    auto   10/100BaseTX
Fa0/17    Fa0/17    notconnect  1       auto    auto   10/100BaseTX
Fa0/18    Fa0/18    notconnect  1       auto    auto   10/100BaseTX
Fa0/19    Fa0/19    notconnect  1       auto    auto   10/100BaseTX
Fa0/20    Fa0/20    notconnect  1       auto    auto   10/100BaseTX
Fa0/21    Fa0/21    notconnect  1       auto    auto   10/100BaseTX
--More--
```



```
COM5 - PuTTY
Switch>
Switch>en
Switch#show interface status

Port      Name      Status      Vlan    Duplex  Speed  Type
Fa0/1     Fa0/1     connected   1       a-full  a-100  10/100BaseTX
Fa0/2     Fa0/2     notconnect  1       auto    auto   10/100BaseTX
Fa0/3     Fa0/3     notconnect  1       auto    auto   10/100BaseTX
Fa0/4     Fa0/4     notconnect  1       auto    auto   10/100BaseTX
```

```
COM5 - PuTTY
Switch#show mac address table
^
% Invalid input detected at '^' marker.

Switch#show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
All     0100.0ccc.cccc    STATIC  CPU
All     0100.0ccc.cccd    STATIC  CPU
All     0180.c200.0000    STATIC  CPU
All     0180.c200.0001    STATIC  CPU
All     0180.c200.0002    STATIC  CPU
All     0180.c200.0003    STATIC  CPU
All     0180.c200.0004    STATIC  CPU
All     0180.c200.0005    STATIC  CPU
All     0180.c200.0006    STATIC  CPU
All     0180.c200.0007    STATIC  CPU
All     0180.c200.0008    STATIC  CPU
All     0180.c200.0009    STATIC  CPU
All     0180.c200.000a    STATIC  CPU
All     0180.c200.000b    STATIC  CPU
All     0180.c200.000c    STATIC  CPU
All     0180.c200.000d    STATIC  CPU
All     0180.c200.000e    STATIC  CPU
All     0180.c200.000f    STATIC  CPU
--More--
```

Verification – MAC Learning

Command:

show mac address-table

Output:

Vlan	Mac Address	Type	Ports
1	c4da.2604.8923	DYNAMIC	Fa0/1

Conclusion:

The switch successfully learned the MAC address of the connected host.
Layer 2 connectivity verified.

```
COM5 - PuTTY
-----
Vlan    Mac Address      Type    Ports
----    -
All     0100.0ccc.cccc    STATIC  CPU
All     0100.0ccc.cccd    STATIC  CPU
All     0180.c200.0000    STATIC  CPU
All     0180.c200.0001    STATIC  CPU
All     0180.c200.0002    STATIC  CPU
All     0180.c200.0003    STATIC  CPU
All     0180.c200.0004    STATIC  CPU
All     0180.c200.0005    STATIC  CPU
All     0180.c200.0006    STATIC  CPU
All     0180.c200.0007    STATIC  CPU
All     0180.c200.0008    STATIC  CPU
All     0180.c200.0009    STATIC  CPU
All     0180.c200.000a    STATIC  CPU
All     0180.c200.000b    STATIC  CPU
All     0180.c200.000c    STATIC  CPU
All     0180.c200.000d    STATIC  CPU
All     0180.c200.000e    STATIC  CPU
All     0180.c200.000f    STATIC  CPU
All     0180.c200.0010    STATIC  CPU
All     ffff.ffff.ffff    STATIC  CPU
1       c4da.2604.8923    DYNAMIC  Fa0/1
Total Mac Addresses for this criterion: 21
Switch#
Switch#
Switch#
```



```
Switch#show mac address-table interface fa0/1
Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       c4da.2604.8923   DYNAMIC Fa0/1
Total Mac Addresses for this criterion: 1
Switch#
```

```
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport port-security
Switch(config-if)#switchport port-security maximum 1
Switch(config-if)#switchport port-security violation shutdown
Switch(config-if)#switchport port-security mac-address sticky
Switch(config-if)#end
Switch#
*Mar 1 01:24:22.684: %SYS-5-CONFIG_I: Configured from console by console
```

Español English Meaning

maximum 1 **maximum 1** Only 1 MAC address allowed on the port

sticky **sticky** Learns MAC automatically and saves it in the running config

violation shutdown **violation shutdown** If another MAC appears → port is **disabled (err-disabled)**

```
COM5 - PuTTY
Switch#
Switch#
Switch#
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport port-security
Switch(config-if)#switchport port-security maximum 1
Switch(config-if)#switchport port-security violation shutdown
Switch(config-if)#switchport port-security mac-address sticky
Switch(config-if)#end
Switch#
*Mar 1 01:24:22.684: %SYS-5-CONFIG_I: Configured from console by console
Switch#show port-security interface fa0/1
Port Security          : Enabled
Port Status            : Secure-up
Violation Mode         : Shutdown
Aging Time             : 0 mins
Aging Type             : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses  : 1
Total MAC Addresses    : 1
Configured MAC Addresses : 0
Sticky MAC Addresses   : 1
Last Source Address:Vlan : c4da.2604.8923:1
Security Violation Count : 0
Switch#
```

- **Enabled** → Port Security active
- **Secure-up** → Port operating normally
- **Violation Mode: Shutdown** → Unauthorized MAC disables the port
 - **Maximum MAC Addresses: 1** → Only one device allowed
 - **Sticky MAC: 1** → MAC learned and stored automatically
- **Violation Count: 0** → No violations detected

```

COM5 - PuTTY
Fa0/24      notconnect  1      auto    auto 10/100BaseTX
Gi0/1       notconnect  1      auto    auto 10/100/1000BaseTX
Gi0/2       notconnect  1      auto    auto 10/100/1000BaseTX
Switch#
Switch#show port-security interface fa0/1
Port Security      : Enabled
Port Status        : Secure-up
Violation Mode      : Shutdown
Aging Time         : 0 mins
Aging Type         : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses : 1
Total MAC Addresses : 1
Configured MAC Addresses : 0
Sticky MAC Addresses : 1
Last Source Address:Vlan : c4da.2604.8923:1
Security Violation Count : 0

Switch#clear port-security sticky interface fa0/1
Switch#clear mac address-table dynamic interface fa0/1
Switch#
*Mar 1 01:42:42.606: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
*Mar 1 01:42:42.606: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to down
*Mar 1 01:42:43.613: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to down
*Mar 1 01:42:59.090: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 01:43:00.097: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

```

REFERENCIA DE COMANDOS / COMMAND REFERENCE

COMANDO / COMMAND	SIGNIFICADO (ES)	MEANING (EN)
<code>switchport port-security</code>	Activa Port Security en el puerto	Enables Port Security on the port
<code>maximum 1</code>	Solo 1 dispositivo permitido por puerto	Only 1 device allowed per port
<code>mac-address sticky</code>	Guarda la MAC aprendida en running-config automáticamente	Saves learned MAC to running-config automatically
<code>violation shutdown</code>	MAC no autorizada -> puerto entra en err-disabled	Unauthorized MAC -> port enters err-disabled state
<code>show port-security interface Fa0/1</code>	Muestra estado completo de Port Security	Shows full Port Security status on the port
<code>show mac address-table</code>	Muestra la tabla MAC dinamica del switch	Displays the switch dynamic MAC address table
<code>shutdown / no shutdown</code>	Ciclo para recuperar puerto en estado err-disabled	Cycle to recover a port from err-disabled state

VERIFICACION — show port-security interface Fa0/1

CAMPO / FIELD	SIGNIFICADO (ES)	MEANING (EN)
Enabled	Port Security activado en el puerto	Port Security activated on the port
Secure-up	Puerto operativo bajo Port Security	Port operating normally under Port Security
Violation Mode: Shutdown	MAC no autorizada -> puerto pasa a err-disabled	Unauthorized MAC -> port enters err-disabled

Maximum MAC Addresses: 1	Solo 1 dispositivo permitido	Only 1 device allowed
Sticky MAC: 1	MAC del equipo guardada en running-config	Device MAC saved to running-config
Violation Count: 0	Sin intentos de acceso no autorizado registrados	No unauthorized access attempts recorded

TROUBLESHOOTING DOCUMENTADO / DOCUMENTED TROUBLESHOOTING

HALLAZGO / FINDING	ACCION CORRECTIVA / CORRECTIVE ACTION
MAC aparece como DYNAMIC antes de Port Security — sin restriccion de acceso en estado inicial. / MAC appears as DYNAMIC before Port Security — no access restriction in initial state.	Baseline documentado con show mac address-table antes de cualquier cambio. Conectividad L2 confirmada. / Baseline documented with show mac address-table before any change. L2 connectivity confirmed.
Con sticky la MAC aprendida se escribe en running-config pero NO en startup-config automaticamente. / With sticky, learned MAC is written to running-config but NOT to startup-config automatically.	Se ejecutó write memory para persistir la configuracion ante reinicios. / Executed write memory to persist configuration across reboots.
Violation Count = 0 al finalizar — no se probó físicamente una segunda MAC en este lab. / Violation Count = 0 at end — second MAC not physically tested in this lab.	Documentado como mejora: conectar segundo dispositivo y verificar transicion a err-disabled + recuperacion con shutdown/no shutdown. / Documented as improvement: connect second device and verify err-disabled transition + recovery.

RESULTADO / RESULT

RESULTADO (ES) Port Security configurado y activo en Fa0/1. MAC sticky aprendida y persistida. Puerto en estado Secure-up con Violation Count 0. Las tres etapas del lab completadas con éxito.	RESULT (EN) Port Security configured and active on Fa0/1. Sticky MAC learned and persisted. Port in Secure-up state with Violation Count 0. All three lab stages completed successfully.
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KEY TAKEAWAY

"En NOC, siempre se documenta el estado normal (baseline) antes de aplicar cambios. Port Security con sticky y violation shutdown es proteccion L2 rapida: si aparece una MAC no autorizada, el puerto entra en err-disabled generando alerta inmediata. El tecnico NOC debe identificar ese estado, investigar la causa y recuperar el puerto con shutdown / no shutdown — sin afectar puertos adyacentes."

"In NOC, the normal state (baseline) is always documented before changes. Port Security with sticky and violation shutdown is fast L2 protection: an unauthorized MAC triggers err-disabled and an immediate alert. The NOC technician must identify the state, investigate, and recover the port with shutdown / no shutdown — without affecting adjacent ports."

NOC RELEVANCE — MINING / INDUSTRIAL ENVIRONMENT

In a mining operation (e.g., Newmont Cerro Negro), unauthorized devices connecting to the network can compromise SCADA systems, IP cameras, or safety communications. Port Security with violation shutdown prevents rogue device insertion at the physical layer. A NOC engineer must: (1) identify err-disabled ports via show interface status, (2) determine cause — unauthorized MAC or cabling error, (3) safely recover the port with shutdown / no shutdown after authorization, and (4) escalate if violation count keeps increasing, which may indicate an active intrusion attempt.